STPM Physics · Semester 2 · Chapter 18

ELECTROMAGNETIC INDUCTION

18.1 Magnetic Flux

DEFINITIONS & LAWS

- **Magnetic flux** through a surface is the product of the magnetic flux density and the area of the surface, measured perpendicular to the field, $\Phi = BA\cos\theta$; it can be pictured as the total number of magnetic field lines passing through the area.
- **Weber (Wb)** is the magnetic flux when a magnetic flux density of 1 T passes perpendicularly through an area of 1 m^2.
- **Magnetic flux linkage** is the product of the number of turns of a coil and the magnetic flux through each turn, $N\Phi$.
- **Magnetic flux density** B is the magnetic flux per unit area perpendicular to the field (the number of field lines per unit normal area), unit tesla (T = Wb m^{-2}).
- For a non-uniform field, the flux is found by summing the normal component of B over all area elements, $\Phi = \int B \cdot dA$.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Magnetic flux	$\Phi = BA\cos\theta$	Φ = magnetic flux (Wb = T m^2); B = magnetic flux density (T); A = area (m^2); θ = angle between B and the normal to the plane
Flux linkage	$N\Phi = NBA\cos\theta$	$N\Phi$ = flux linkage (Wb-turns); N = number of turns
Maximum flux	$\Phi = BA$ at $\theta = 0°$	Plane perpendicular to B
Zero flux	$\Phi = 0$ at $\theta = 90°$	Plane parallel to B
Area of circular coil	$A = \pi r^2$	A = area (m^2); r = radius (m)
Area of square coil	$A = (\text{side})^2$	A = area (m^2); side in m

TIPS TO ANSWER

- State the formula $\Phi = BA\cos\theta$, define θ as the angle between B and the normal to the coil, then substitute with SI units and give the answer in Wb.
- For flux linkage multiply by N and state the unit as Wb turns (or Wb).

- Convert cm^2 to m^2 ($1\ cm^2 = 10^{-4}\ m^2$) and radius to metres before calculating the area.
- In a descriptive answer, state that the flux depends on the field strength, the area and the orientation of the coil relative to the field.

SECTION A TRAPS

Trap: The angle θ is taken between B and the plane of the coil. This is wrong: θ is between B and the normal. A plane parallel to B has zero flux, not maximum flux.

Trap: Using $A = \pi d^2$ with the diameter, or leaving the area in cm^2, gives an answer wrong by a factor of 4 or 10^4.

Trap: Forgetting N when the question asks for flux linkage; the flux through one turn is not the flux linkage.

TRICKS

Trick: Plane perpendicular to B gives the maximum $\Phi = BA$; plane parallel to B gives $\Phi = 0$. Any $\cos\theta$ value between these can be checked as a fraction of BA ($\cos 60° = 0.5$, $\cos 30° = 0.866$).

18.2 Faraday's Law and Lenz's Law

DEFINITIONS & LAWS

- **Faraday's law**: the magnitude of the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage (or the rate of cutting of magnetic flux).
- **Electromagnetic induction** is the production of an induced e.m.f. (and an induced current in a closed circuit) when the magnetic flux linked with a conductor changes.
- **Induced e.m.f.** is the e.m.f. produced in a conductor when there is a change in the magnetic flux linked with it.
- **Lenz's law**: the direction of the induced current is such that it opposes the change in magnetic flux that produces it.
- Lenz's law is a consequence of the conservation of energy: work must be done against the opposing effect to produce the induced current.
- Induced e.m.f. appears whenever the flux linkage changes, by changing B, A or θ, whether or not the circuit is closed; an induced current flows only in a closed circuit.
- For a straight conductor moving across a uniform field, the induced e.m.f. is the product of B, the length of the conductor and the velocity perpendicular to B.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Faraday's law	$\varepsilon = -N\frac{d\Phi}{dt} = -\frac{d(N\Phi)}{dt}$	ε = induced emf (V); N = number of turns; Φ = magnetic flux (Wb); t = time (s); the minus sign represents Lenz's law
Faraday's law, uniform change	$\lvert\varepsilon\rvert = \frac{N\Delta\Phi}{\Delta t}$	ε = induced emf (V); N = number of turns; ΔΦ = change in flux (Wb); Δt = time interval (s); for a uniform change
Emf induced in a moving rod	$\varepsilon = BLv$	ε = emf (V); B = magnetic flux density (T); L = length of rod (m); v = velocity of rod (m s^{-1}); B, L and v mutually perpendicular
Emf with v at angle to B	$\varepsilon = BLv\sin\theta$	θ = angle between v and B; other symbols as above
Rod in a closed circuit	$I = \frac{BLv}{R}$; $F = BIL = \frac{B^2L^2v}{R}$; $P = Fv = \frac{\varepsilon^2}{R}$	I = current (A); R = resistance of circuit (Ω); F = force (N); P = power (W); B, L, v as above; ε = BLv
Emf in a rotating coil	$\varepsilon = NBA\omega\sin\omega t$; $\varepsilon_0 = NBA\omega$	ε = induced emf (V); ε_0 = peak emf (V); N = number of turns; B = magnetic flux density (T); A = area (m^2); ω = angular speed (rad s^{-1}); uniform field

TIPS TO ANSWER

- For a calculation, state $\varepsilon = -N\frac{d\Phi}{dt}$, evaluate ΔΦ (final minus initial, with area and angle), divide by Δt, and give the magnitude in V with unit.
- For direction, apply Lenz's law in three steps: state the change in flux (increase or decrease), state that the induced field opposes this change, then deduce the current direction (right-hand grip rule) or the pole formed.
- Pole rule: an approaching pole induces the same pole at the near end of the coil (repulsion); a receding pole induces the opposite pole (attraction).
- In 'explain why' answers, mention the conservation of energy: if the induced current assisted the change, energy would be created.
- A copper plate falling past a strong magnet: the changing flux induces eddy currents whose magnetic force opposes the motion (Lenz's law), so its acceleration is less than g.
- When asked for the external force needed to keep a rod moving at constant speed, give its value: $P = \frac{\varepsilon^2}{R} = Fv$, so $F = B^2I^2v/R$ (not just "push" or "pull").

SECTION A TRAPS

Trap: Writing that the induced e.m.f. is proportional to the magnetic flux. It is proportional to the rate of change of flux; a large but constant flux gives zero e.m.f.

Trap: Stating that the induced current opposes the magnetic field or the flux. It opposes the change in flux; when the flux decreases, the induced field is in the same direction as the original field.

Trap: Forgetting the factor N for a coil, or using the final flux instead of the change in flux, $\Delta\Phi$.

Trap: Doubling v in a rod and rails circuit doubles ε and I, but the power $\frac{\varepsilon^2}{R}$ and the force on the rod change by different factors: P becomes 4 times, F becomes 2 times.

TRICKS

Trick: Because $P = \frac{\varepsilon^2}{R}$ and $\varepsilon \propto v$, any change in speed by a factor k changes the power by k^2, while the retarding force changes by k. Check by ratio instead of recalculating.

18.3 Self Induction

DEFINITIONS & LAWS

- **Self-induction** is the production of an induced e.m.f. in a coil due to a change in the current in the same coil, which changes the magnetic flux linked with it.
- **Self-inductance** L of a coil is the ratio of the magnetic flux linkage in the coil to the current flowing through it, $L = \frac{N\Phi}{I}$.
- **Henry (H)** is the self-inductance of a coil in which an e.m.f. of 1 V is induced when the current changes at the rate of 1 A s^{-1}.
- Self-inductance can also be defined as the back e.m.f. induced in a coil per unit rate of change of current in the coil, $L = -\varepsilon/\frac{dI}{dt}$.
- **Back e.m.f.** is the induced e.m.f. produced in a circuit by a change in its own current; it opposes the change in current that produces it (Lenz's law).
- By Lenz's law, the self-induced e.m.f. always opposes the change in current: it opposes an increase in current and opposes a decrease in current.
- The self-induced e.m.f. is large when the current is switched off very quickly, since $\frac{dI}{dt}$ is large; this explains the spark across the switch contacts.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Self-induced emf	$\varepsilon = -L\frac{dI}{dt}$	ε = induced emf (V); L = self-inductance (H); $\frac{dI}{dt}$ = rate of change of current (A s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Self-induced emf, uniform change	$\lvert\varepsilon\rvert = L\,\frac{\Delta I}{\Delta t}$	ΔI = change in current (A); Δt = time interval (s); for a uniform change
Self-inductance	$L = \frac{N\Phi}{I}$	L = self-inductance (H = Wb A^{-1} = V s A^{-1}); N = number of turns; Φ = flux (Wb); I = current (A)
Inductance of air-cored solenoid	$L = \frac{\mu_0 N^2 A}{l}$	L = self-inductance (H); N = number of turns; A = cross-sectional area (m^2); l = length (m); μ_0 = $4\pi\times10^{-7}$ H m^{-1}; long air-cored solenoid
Inductance with a core	$L = \frac{\mu_r \mu_0 N^2 A}{l}$	μ_r = relative permeability of the core (no unit); other symbols as above

TIPS TO ANSWER

- Define L using flux linkage and current $\frac{N\Phi}{I}$, not e.m.f. and current alone; you may add that $\varepsilon = -L\,\frac{dI}{dt}$.
- For a calculation, state $\lvert\varepsilon\rvert = L\,\frac{\Delta I}{\Delta t}$, take ΔI as the magnitude of the change in current (final minus initial), substitute with units and give the answer in V.
- In an explanation of the delay in the growth of current, state: the changing current induces an e.m.f. which opposes the change, so the current rises or falls gradually and not instantly.
- For $L = \frac{\mu_0 N^2 A}{l}$, always state which quantity changes and apply the proportionality: $L \propto N^2$, $L \propto A$, $L \propto \frac{1}{l}$.

SECTION A TRAPS

Trap: Using $L \propto N$ instead of $L \propto N^2$. Doubling the turns on a fixed length gives 4 times the inductance.

Trap: Stretching the solenoid to twice the length with N and A fixed halves L, since $L \propto \frac{1}{l}$; it does not double it.

Trap: Stating that the self-induced e.m.f. opposes the current. It opposes the change in current: it opposes an increase and, when the current decreases, it acts in the same direction as the current to oppose the decrease.

Trap: Taking $\Delta I = 0$ because the final current is zero; ΔI is final minus initial, for example 6.0 − 2.0 = 4.0 A, not 6.0 A.

TRICKS

> **Trick:** Check unit consistency: H = V s A^{-1}, so e.m.f. = H × A s^{-1} gives V automatically; if the answer comes out in mV or kV, check the powers of ten in Δt.

18.4 Energy Stored in an Inductor

DEFINITIONS & LAWS

- The energy stored in an inductor is the work done against the self-induced e.m.f. in establishing the current; it is stored in the magnetic field of the inductor.
- The energy is released when the current decreases, as the self-induced e.m.f. then maintains the current for a short time.
- The energy stored per unit volume of a magnetic field is the energy density, $u = \frac{B^2}{2\mu_0}$.
- Energy is stored in the magnetic field of the inductor, not lost; in an ideal inductor there is no energy dissipation.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Energy stored in an inductor	$E = \frac{1}{2}LI^2$	E = energy (J); L = self-inductance (H); I = current (A)
Energy in terms of flux linkage	$E = \frac{N\Phi I}{2}$	N = number of turns; Φ = flux (Wb); I = current (A); since NΦ = LI
Magnetic energy density	$u = \frac{B^2}{2\mu_0}$	u = energy density (J m^{-3}); B = magnetic flux density (T); $\mu_0 = 4\pi\times10^{-7}$ H m^{-1}
Average power released	$P = \frac{E}{t}$	P = average power (W); E = energy (J); t = time (s)

TIPS TO ANSWER

- State $E = \frac{1}{2}LI^2$, substitute with SI units (L in H, I in A) and give the answer in J with 3 s.f.
- For a current changing from I_1 to I_2, calculate the change in energy as $\frac{1}{2}L(I_2^2 - I_1^2)$; do not square the difference of the currents.
- To find I or L from a given energy, rearrange first: $I = \sqrt{\frac{2E}{L}}$ and $L = \frac{2E}{I^2}$.
- For power over a time interval, state $P = \frac{E}{t}$ with the energy released in that time.

SECTION A TRAPS

Trap: Using $E = LI^2$ or $E = \frac{1}{2}LI$ (missing the factor ½ or the square), or $E = \frac{1}{2}L^2I$.

Trap: Squaring after doubling incorrectly: doubling I multiplies E by 4, and tripling I multiplies E by 9, but doubling L only doubles E.

Trap: In an energy-density question, forgetting the factor 2 in the denominator, or taking $\mu_0 = 4\pi\times10^{-7}$ without the 4π.

TRICKS

Trick: Use proportion: $E \propto L$ at fixed I and $E \propto I^2$ at fixed L. For example, I tripled gives 9 times the energy; compare options by ratio before evaluating.

18.5 Mutual Induction

DEFINITIONS & LAWS

- **Mutual induction** is the production of an induced e.m.f. in one coil due to a change in the current in a neighbouring coil, which changes the magnetic flux linked with the first coil.
- **Mutual inductance** M is the ratio of the magnetic flux linkage in one coil to the current in the other coil that produces the flux, $M = \frac{N_2\Phi_{21}}{I_1}$.
- **Henry (H)** is the mutual inductance of two coils when a current changing at the rate of 1 A s^{-1} in one coil induces an e.m.f. of 1 V in the other.
- Mutual inductance can also be defined as the e.m.f. induced in one coil per unit rate of change of current in the other coil, $M = -\varepsilon_2/\frac{dI_1}{dt}$.
- Mutual inductance depends on the geometry of the coils (number of turns, size, separation, relative orientation) and the permeability of the core, not on the current.
- **Transformer**: an alternating current in the primary coil produces a changing flux in the core, which induces an alternating e.m.f. in the secondary coil by mutual induction.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Mutually induced emf	$\varepsilon_2 = -M\frac{dI_1}{dt}$	ε_2 = emf induced in secondary (V); M = mutual inductance (H); $\frac{dI_1}{dt}$ = rate of change of primary current (A s^{-1})
Mutually induced emf, uniform change	$\lvert\varepsilon_2\rvert = M\frac{\Delta I_1}{\Delta t}$	ΔI_1 = change in primary current (A); Δt = time interval (s); for a uniform change

Quantity / law	Formula	Symbols, units & conditions
Mutual inductance	$N_2\Phi_{21} = MI_1$	N_2 = number of turns of secondary; Φ_{21} = flux in secondary due to primary (Wb); M = mutual inductance (H = Wb A^{-1}); I_1 = primary current (A)
Ideal transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$; $V_pI_p = V_sI_s$	V = voltage (V); N = number of turns; I = current (A); subscripts p = primary, s = secondary
Coupling coefficient	$M = k\sqrt{L_1L_2}$	M = mutual inductance (H); k = coupling coefficient, $0 \le k \le 1$; L_1, L_2 = self-inductances of the two coils (H)

TIPS TO ANSWER

- Define M with the flux linkage in one coil and the current in the other coil (cross-reference of two coils), not the same coil.
- State $\varepsilon = -M\frac{dI}{dt}$, substitute M and the rate of change of current $\frac{\Delta I}{\Delta t}$ with units, and give the answer in V.
- For a transformer, state that the changing flux in the core links both coils, and that $\frac{V_s}{V_p} = \frac{N_s}{N_p}$; identify step-up ($N_s$ greater than N_p) or step-down.
- Explain that a steady direct current in the primary induces no e.m.f. in the secondary because the flux does not change.
- Explain mutual induction in sequence: current in the primary coil produces a magnetic flux; the flux links the secondary coil; when the primary current changes, the flux linkage in the secondary changes; an e.m.f. is induced in the secondary (Faraday's law).

SECTION A TRAPS

Trap: Writing that M depends on the magnitude of the primary current. It does not; only the induced e.m.f. depends on the rate of change of current.

Trap: A transformer ratio inverted: $V_s = V_p \times \frac{N_p}{N_s}$. The secondary voltage is directly proportional to the secondary turns.

Trap: Believing that a constant rate of change of current gives an increasing induced e.m.f. If $\frac{dI}{dt}$ is constant, ε_2 is constant.

Trap: Not converting mH to H (25 mH = 25×10^{-3} H) or ms to s.

TRICKS

> **Trick:** For a transformer, check the sign of the answer by comparing turns: fewer secondary turns means a lower voltage. A 20:1 turns ratio on 240 V gives 12 V.